contains a particular rare gene that I contain, and the chances are equally good that your sister contains a rare gene in common with you. The odds are in this case exactly 50 per cent, and it is easy to explain why.

Suppose you contain one copy of the gene *G*. You must have received it either from your father or from your mother (for convenience we can neglect various infrequent possibilities—that *G* is a new mutation, that both your parents had it, or that either of your parents had two copies of it). Suppose it was your father who gave you the gene. Then every one of his ordinary body cells contained one copy of *G*. Now you will remember that when a man makes a sperm he doles out half his genes to it. There is therefore a 50 per cent chance that the sperm that begot your sister received the gene *G*. If, on the other hand, you received *G* from your mother, exactly parallel reasoning shows that half of her eggs must have contained *G*; once again, the chances are 50 per cent that your sister contains *G*. This means that if you had 100 brothers and sisters, approximately 50 of them would contain any particular rare gene that you contain. It also means that if you have 100 rare genes, approximately 50 of them are in the body of any one of your brothers or sisters.

You can do the same kind of calculation for any degree of kinship you like. An important relationship is that between parent and child. If you have one copy of gene *H*, the chance that any particular one of your children has it is 50 per cent, because half your sex cells contain *H*, and any particular child was made from one of those sex cells. If you have one copy of gene *J*, the chance that your father also had *J* is 50 per cent, because you received half your genes from him, and half from your mother. For convenience we use an index of *relatedness*, which expresses the chance of a gene being shared between two relatives. The relatedness between two brothers is $\frac{1}{2}$, since half the genes possessed by one brother will be found in the other. This is an average figure: by the luck of the meiotic draw, it is possible for particular pairs of brothers to share more or fewer genes than this. The relatedness between parent and child is always exactly $\frac{1}{2}$.

It is rather tedious going through the calculations from first principles every time, so here is a rough and ready rule for working out the relatedness between any two individuals *A* and *B*. You may find it useful in making your will, or in interpreting apparent resemblances in your